

3 point

Which type of environment are most suitable for AND-OR trees?

- ☐ Unknown environments
- ☐ Nondeterministic environments
- ☐ Online environments

☒ Non-observable environments

3 point

What is a key characteristic of 'general AI' as discussed in Chapter 28 Philosophy, Ethics, and Safety of AI?

- ☐ Limited to predefined functions
 - ☒ Capable of performing any intellectual task that a human can do
 - ☐ Dependent on specific programming
 - ☐ Specialized in narrow tasks
-

Which algorithm extends the minimax approach to handle games involving elements of chance?

☒ Expectiminimax algorithm

☐ A*

☐ Monte Carlo Tree Search

☐ Alpha-beta pruning

What does the soundness property ensure in propositional logic?

☒ If a sentence can be derived using inference rules, then it must be true in all models.

☐ Any two logically equivalent sentences must have the same truth value.

☐ Every logical system has at least one model.

☐ If a sentence is true in all models, then it can be derived using inference rules.

3 point

In First-Order Logic, what does the expression $\forall x (P(x) \vee Q(x))$ mean precisely?

- ☐ For all x , $P(x)$ and $Q(x)$ are both true.
 - ☐ There exists an x for which either $P(x)$ or $Q(x)$ is true.
 - ☒ For all x , either $P(x)$ is true or $Q(x)$ is true (or both).
 - ☐ There exists an x for which $P(x)$ and $Q(x)$ are both true.
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3 point

For a self-driving car, which of the following is considered a sensor?

- ☒ Speedometer.
 - ☐ Steering wheel.
 - ☐ Engine.
 - ☐ Brake pedal.
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3 point

What is the primary difference between informed and uninformed search strategies?

- ☐ Uninformed strategies guarantee optimal solutions.
- ☐ Uninformed strategies require less memory.

☒ Informed strategies use heuristics to guide the search.

☐ Informed strategies explore all possible nodes.

3 point

For which type of games is the minimax algorithm particularly well-suited?

☒ Two-player, zero-sum, turn-based games with perfect information

☐ Single-player puzzle games

☐ Real-time strategy games

☐ Multiplayer games with hidden information

What does the existential quantifier (*exists*) signify in First-Order Logic?

- ☐ That a statement is true for all possible instances.
 - ☐ That a statement is true and false simultaneously.
 - ☒ That there exists at least one instance for which the statement is true.
 - ☐ That a statement is false for all instances.
-

Which of the following is a key feature of the Planning Domain Definition Language (PDDL)?

- ☐ It is limited to representing only stochastic planning problems.
 - ☒ It is designed to standardize the representation of planning problems across different planners.
 - ☐ It does not support the definition of action preconditions and effects.
 - ☐ It is primarily used for representing neural network configurations.
-

Which of the following statements is true?

- ☐ Every contradiction is satisfiable.
- ☐ No tautology is satisfiable.
- ☐ Every satisfiable statement is a tautology.

☒ Every tautology is satisfiable.

Which of the following represents the logical AND operation?

☒ $\wedge$

☐ $\neg$

☐ $\vee$

☐ $\rightarrow$

What is the primary goal of the Turing Test in the context of AI?

- ☒ To evaluate a machine's ability to exhibit intelligent behavior indistinguishable from a human
 - ☐ To assess a machine's ability to solve mathematical problems
 - ☐ To measure a machine's processing speed
 - ☐ To determine a machine's capacity for learning
-

Which of the following environments is both deterministic and fully observable?

- ☐ Self-driving car navigation
- ☐ Stock market
- ☐ Poker game

- ☒ Chess game

The "Disability Argument" in artificial intelligence fundamentally challenges the notion that:

- ☐ AI systems will inevitably surpass human intelligence.
 - ☐ Machine learning can improve without explicit programming.
 - ☒ AI can fully replicate uniquely human traits such as creativity and emotions.
 - ☐ Machines can simulate human cognitive processes effectively.
-

Which of the following best describes the sum rule (addition rule) in probability theory?

- ☐ It calculates the probability of the intersection of two events.
 - ☐ It determines the probability of the union of two events.
 - ☒ It finds the probability of two independent events occurring together.
 - ☐ It computes the probability of the complement of an event.
-

Which of the following environments is considered entirely episodic?

- ☐ Teaching a class
 - ☐ Stock market trading
 - ☒ Image classification tasks
 - ☐ Autonomous driving
-

What would be an appropriate performance measure for an AI designed to play a strategy game?

- ☐ The speed at which the AI makes decisions.
 - ☒ The win/loss ratio over multiple games.
 - ☐ The complexity of the algorithms used.
 - ☐ The graphical quality of the game interface.
-

Which of the following is a limitation of the standard minimax algorithm?

- ☐ It is inefficient for games with large search trees due to exhaustive exploration.
- ☐ It is only applicable to games with imperfect information.
- ☐ All of the other answers

☒ It can not easily be adapted to handle games with more than two players.

What is the difference between syntax and semantics?

- ☐ Semantics focuses on sentence formation rules, and syntax determines truth values.
- ☐ Syntax defines the structure and formation rules of sentences, while semantics assigns meaning by defining the truth of sentences in a given world.
- ☐ Syntax and semantics both define the meaning of logical connectives.

☒ Syntax determines the truth of a sentence in a world, while semantics describes how sentences are formed.

In a Constraint Satisfaction Problem (CSP), what does a 'variable' represent?

- ☐ A constraint that must be satisfied
- ☐ A function defining relationships between elements
- ☐ A possible value that can be assigned

☒ An element that can take on different values

Which of the following formulas represents the sum rule for two events, A and B ?

- ☐ $P(A \cap B) = P(A) \times P(B \mid A)$
- ☐ $P(A \cup B) = P(A) \times P(B)$

☒ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

☐ $P(A \cap B) = P(A) + P(B)$

3 point

What does the 'least-constraining value' heuristic aim to achieve in CSPs?

☐ Selects the value that rules out the most choices for neighboring variables

☒ Selects the value that rules out the fewest choices for neighboring variables

☐ Selects the value that is least frequently used

☐ Selects the value that is most frequently used

3 point

What characterizes a stochastic environment?

☒ The environment has elements of randomness and unpredictability

☐ The environment does not change over time

☐ The next state of the environment is determined by the current state and the agent's action

☐ The environment is completely predictable

Which of the following is NOT a characteristic of a rational agent?

- ☐ Makes decisions based on percept history
- ☐ Acts to achieve the best expected outcome
- ☐ Operates autonomously

☒ Always mimics human behavior

What distinguishes 'classical planning' from other forms of planning in AI?

☒ It assumes a fully observable, deterministic, finite, static, and discrete environment.

- ☐ It incorporates learning mechanisms to adapt plans over time.
 - ☐ It focuses on real-time dynamic planning scenarios.
 - ☐ It deals exclusively with stochastic and partially observable environments.
-

In the PEAS description of an agent, what does 'Actuators' refer to?

- ☐ They process sensory information.
 - ☒ They execute actions that allow the agent to interact with the environment.
 - ☐ They measure the performance of the agent.
 - ☐ They define the environment for the agent.
-

What is the main advantage of using 'heuristic search' in planning algorithms?

- ☐ It eliminates the need for defining action preconditions and effects.
 - ☐ It guarantees finding the optimal plan in all scenarios.
 - ☒ It possibly reduces the search space by providing estimates of the cost to reach the goal, leading to more efficient planning.
 - ☐ It allows for planning in non-deterministic environments without any modifications.
-

3 point

How would you represent the statement "There exists a person who is a student and is studying Computer Science" in First-Order Logic?

- ☐ $\forall x (\text{Person}(x) \wedge \text{Student}(x) \wedge \text{Studies}(x, \text{ComputerScience}))$
- ☐ $\exists x (\text{Person}(x) \vee \text{Student}(x) \vee \text{Studies}(x, \text{ComputerScience}))$
- ☐ $\forall x (\text{Person}(x) \rightarrow \text{Student}(x) \rightarrow \text{Studies}(x, \text{ComputerScience}))$
- ☒ $\exists x (\text{Person}(x) \wedge \text{Student}(x) \wedge \text{Studies}(x, \text{ComputerScience}))$

3 point

In hierarchical planning, what is the purpose of 'task decomposition'?

- ☐ To eliminate unnecessary tasks from the plan.
- ☐ To sequence tasks in chronological order.
- ☐ To assign tasks to different agents in a multi-agent system.
- ☒ To break down complex tasks into simpler, more manageable subtasks.